LIVING MORPHOLOGIES

PHYSARUM LOGIC › **OPTIMIZATION** › VERTICAL PROPAGATION › HYBRID CONNECTION

EDIT

PROCESS CATALOG

01

PHYSARUM LOGIC

ARCHITECTURAL INPUT

VERTICAL VOID · FOUR GENERATIONS, THEN ONE MORPHOLOGY FOR VERTICAL PROPAGATION

LOBBY → **VERTICAL VOID**

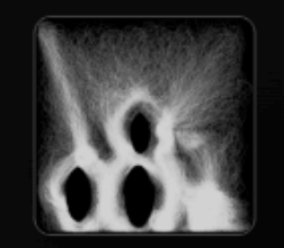
Three categories of analysis. Formal, Spatial, and Atmospheric are what the translation receives.

FORMAL	DYNAMIC CORE	HIGH
SPATIAL	OPEN THRESHOLD	MEDIUM
ATMOSPHERIC	VISUAL IMMERSION	HIGH

BIOLOGICAL TRANSLATION

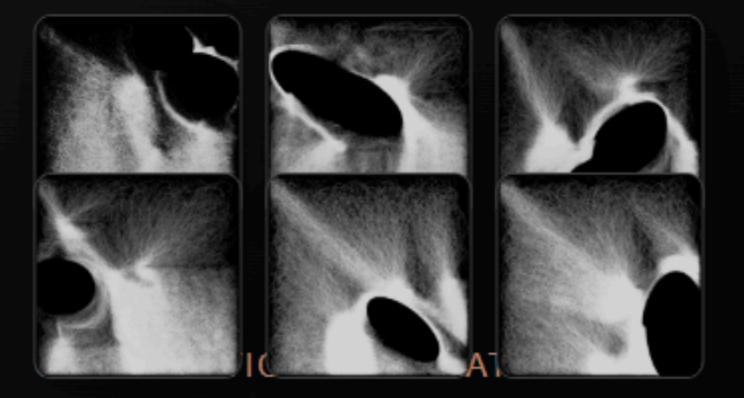
The ratings become Physarum behaviour. This field is one drawing from the Vertical Void catalog.

VERTICAL VOID



CATALOG

Iterations from the Vertical Void catalog. Generation 01 opens from this kind of morphology.



02

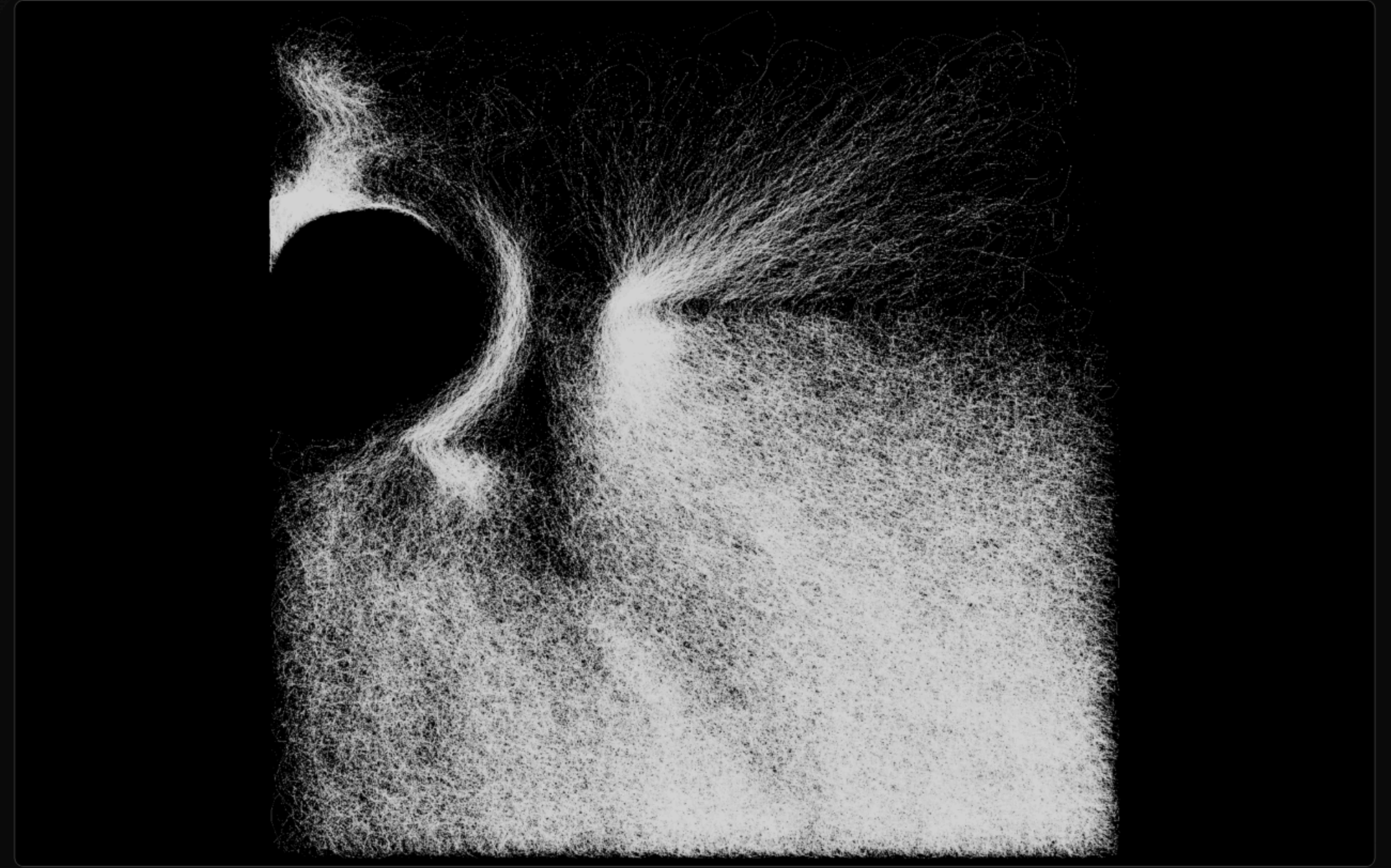
EVOLUTIONARY PROCESS

VERTICAL VOID

OPEN THE SPACE

GENERATION 01

Nothing has been scored, so every new candidate is an explorer from the Physarum translation.



EXPLORERS 100%

NON-DOMINATED —

DIVERSITY —

BEGIN REFINEMENT

GENERATION 02

Most candidates still explore. A minority are children of non-dominated parents, and a smaller share come from diversity parents.

EXPLORERS 70%

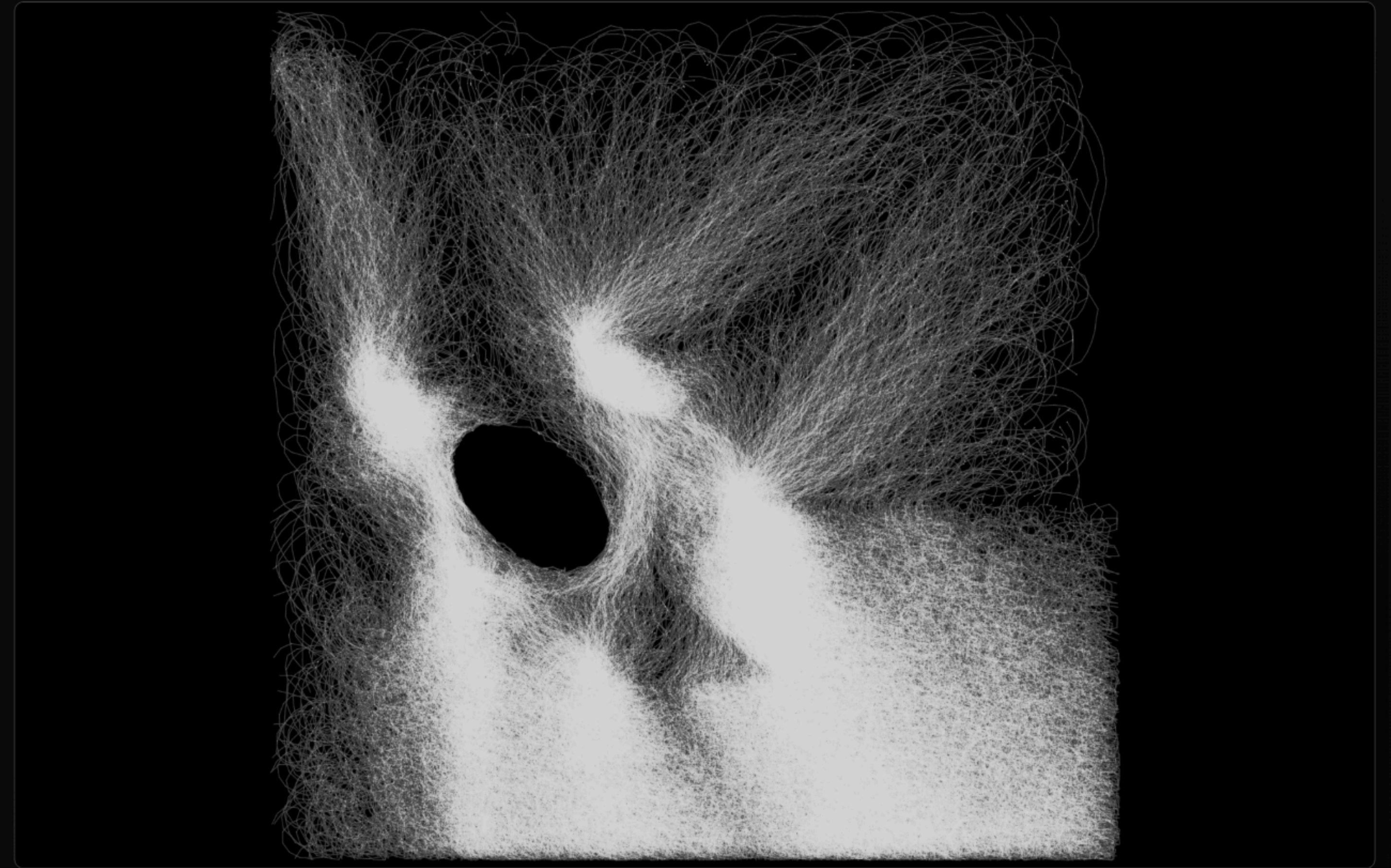
NON-DOMINATED 22%

DIVERSITY 8%

SHIFT THE WEIGHT

GENERATION 03

Explorers and children of kept morphologies share the population. Non-dominated parents are the larger inherited share.



EXPLORERS 50%

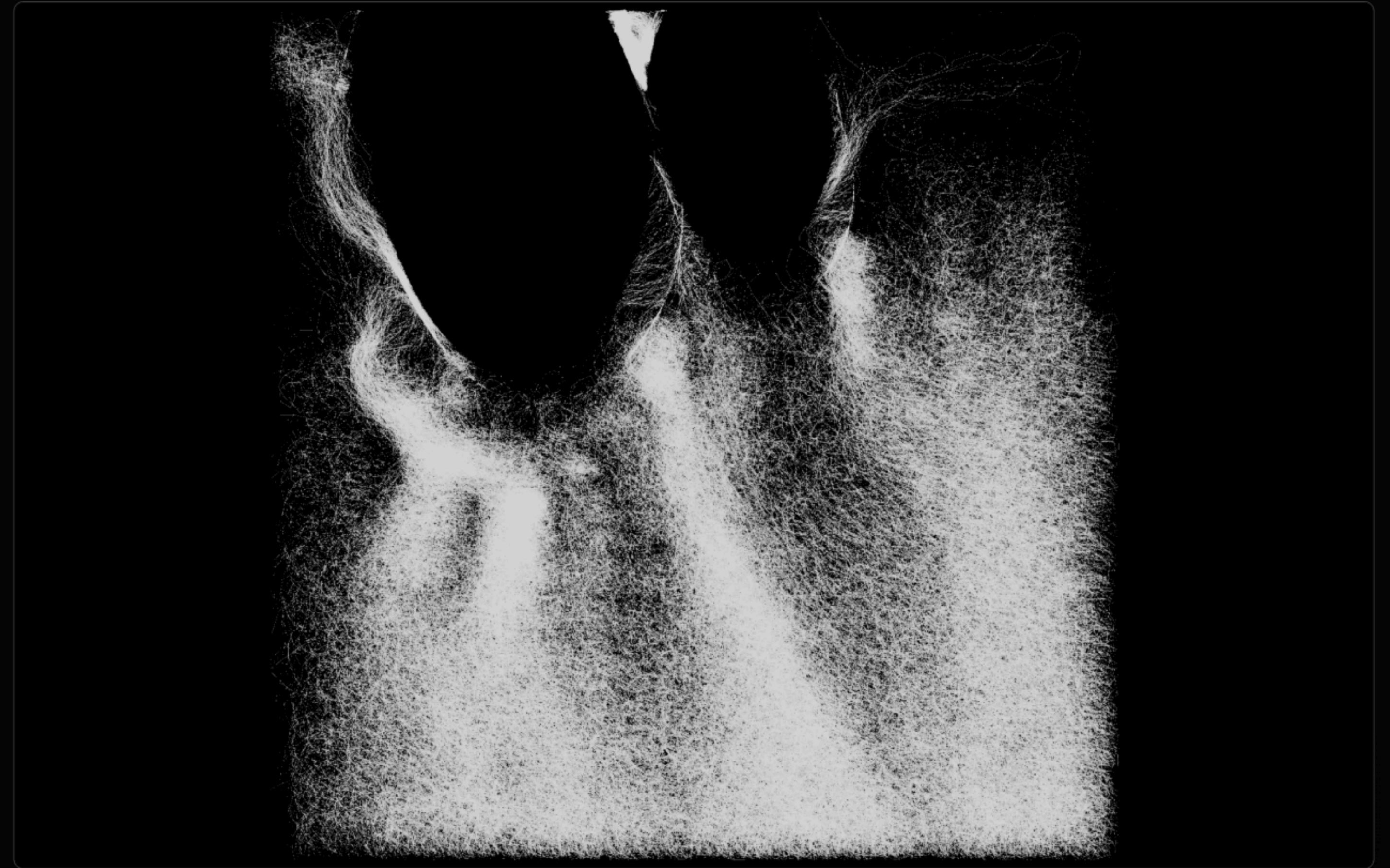
NON-DOMINATED 37%

DIVERSITY 13%

CONCENTRATE

GENERATION 04

Children of non-dominated parents become the majority. Explorers and diversity parents remain. The search then stops.



EXPLORERS 30%

NON-DOMINATED 52%

DIVERSITY 18%

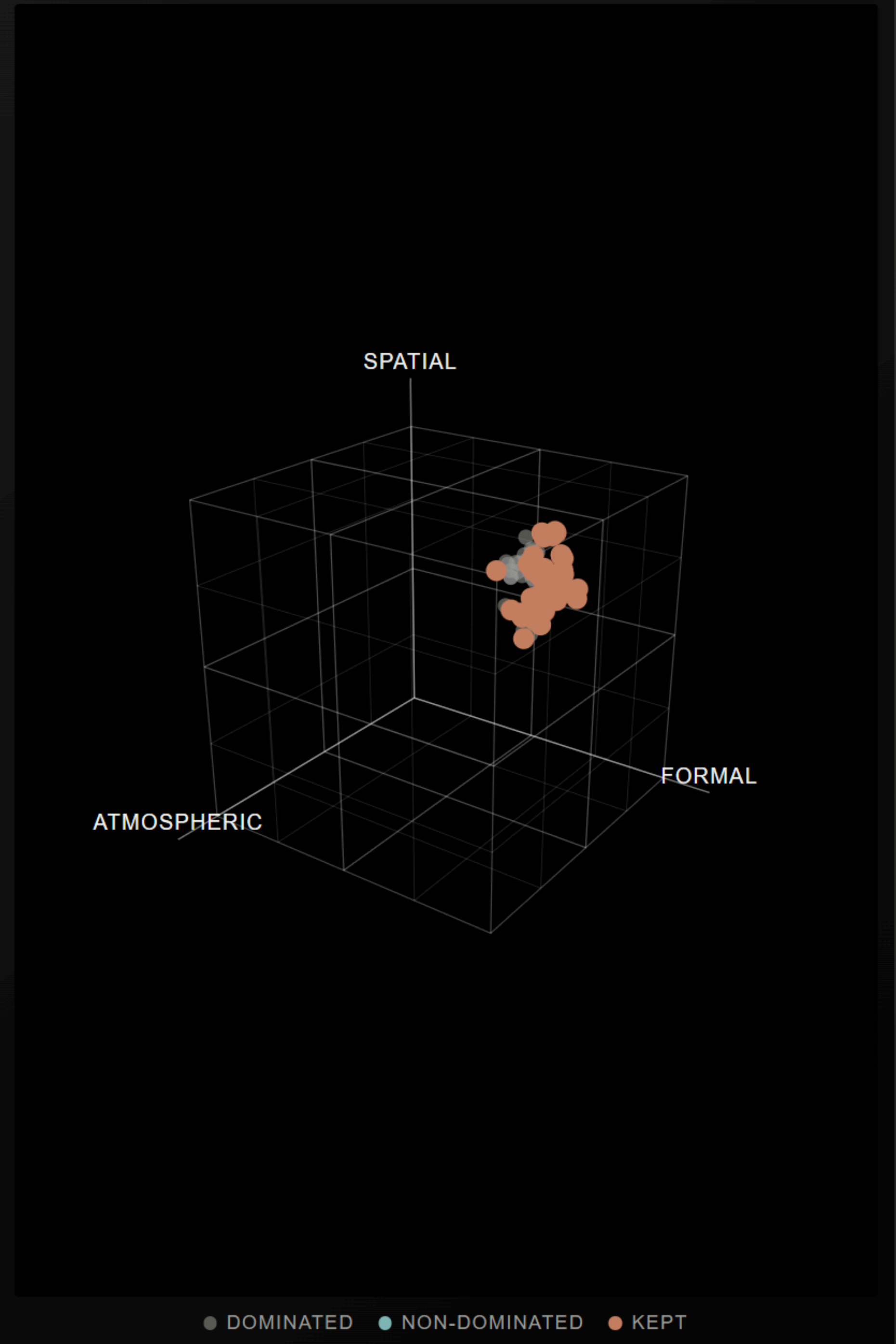
03

OPTIMIZATION OUTPUT

The Pareto front, the roles that keep a morphology, and the human selection.

PARETO

The non-dominated set, continued on the Pareto tab.

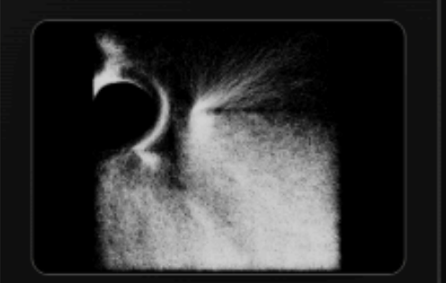


PRESERVATION ROLES

Specialists are off. A morphology is kept for one of these roles.

NON-DOMINATED

Nothing else is at least as strong on all three objectives and stronger on one.



MORPHOLOGICAL DIVERSITY

The phenotype sits apart from the morphologies already kept, so the catalog does not collapse to one shape.

HUMAN SELECTION

Saved drawings, one for each archetype, with the scores from the search.